

Multi-class Perceptron for Simpsons Character Classification

Grayscale and RGB comparison, model selection, and augmentation

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10,000

images per modality

10

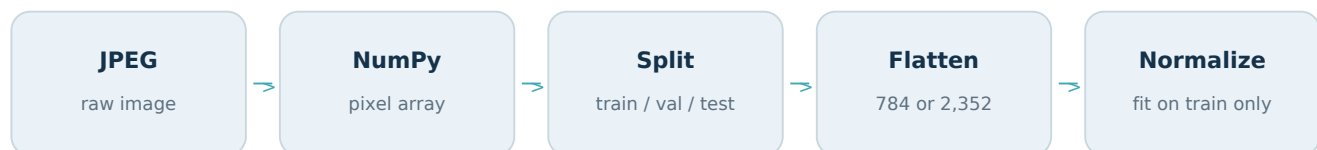
one-vs-rest classifiers

42.85%

best RGB test accuracy

1 Problem and data pipeline

The objective was to implement a perceptron from scratch and classify ten Simpsons characters. The supplied data contained 8,000 training and 2,000 test images in both grayscale and RGB. A stratified split divided the training portion into 6,400 training and 1,600 validation images; the provided test set remained untouched during model selection.



2 Object-oriented model design

BinaryPerceptron

- Own weight vector and bias, both initialized to 0.1.
- Computes a linear score and thresholded binary prediction.
- Updates one image at a time from the prediction error.

MultiClassPerceptron

- Contains ten independent binary classifiers.
- Uses one-vs-rest targets for each true class.
- Returns the class with the largest raw score using argmax.

Online perceptron update

$$w_{\text{new}} = w_{\text{old}} + \eta e x \quad | \quad b_{\text{new}} = b_{\text{old}} + \eta e \quad | \quad \text{score}_k = w_k \text{ dot } x + b_k$$

3 Training and stopping criteria

Every epoch shuffled the 6,400 training images and applied online updates. Two stopping rules were compared on the same grayscale baseline (0-1 scaling, eta = 0.01). Early stopping monitored validation accuracy with patience 3 and restored the best weights.

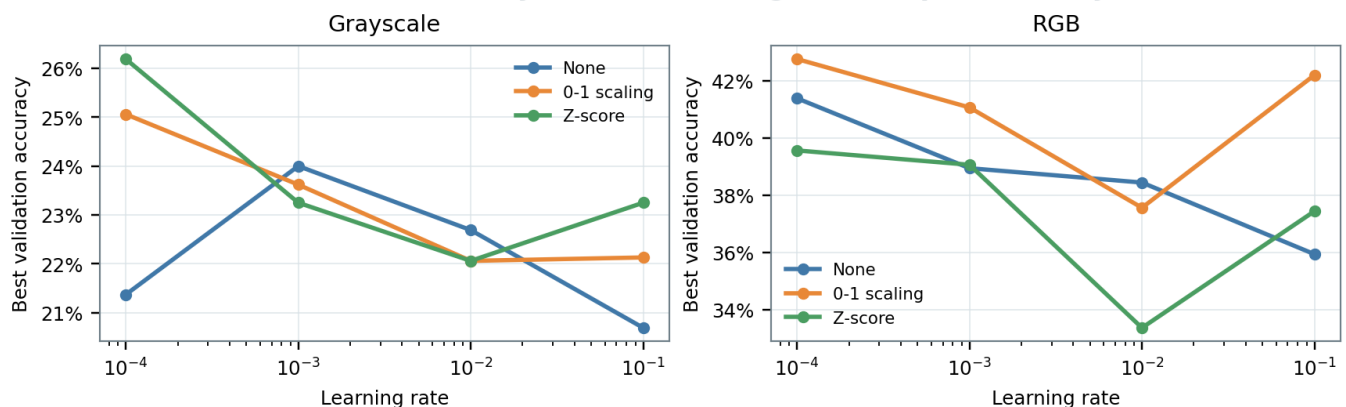
Stopping rule	Epochs run	Best epoch	Best val. accuracy	Time (s)
Fixed 10 epochs	10	10	21.31%	3.02
Early stopping	9	6	21.06%	2.72

Interpretation. Fixed training achieved a slightly higher validation score (+0.25 percentage points), while early stopping used 10% less time and returned the best observed epoch. Because the difference is small, early stopping was used for tuning to control computation and avoid retaining later, non-improving weights.

4 Hyperparameter tuning

For each modality, 12 configurations combined four learning rates (10^{-4} , 10^{-3} , 10^{-2} , 10^{-1}) with no normalization, 0-1 scaling, or z-score normalization. Selection used validation accuracy only. Normalization parameters were estimated from training data and then applied unchanged to validation and test data.

Validation accuracy across 12 configurations per modality



Selected grayscale model

Z-score | eta = 0.0001

Validation accuracy: 26.19%

Selected RGB model

0-1 scaling | eta = 0.0001

Validation accuracy: 42.75%

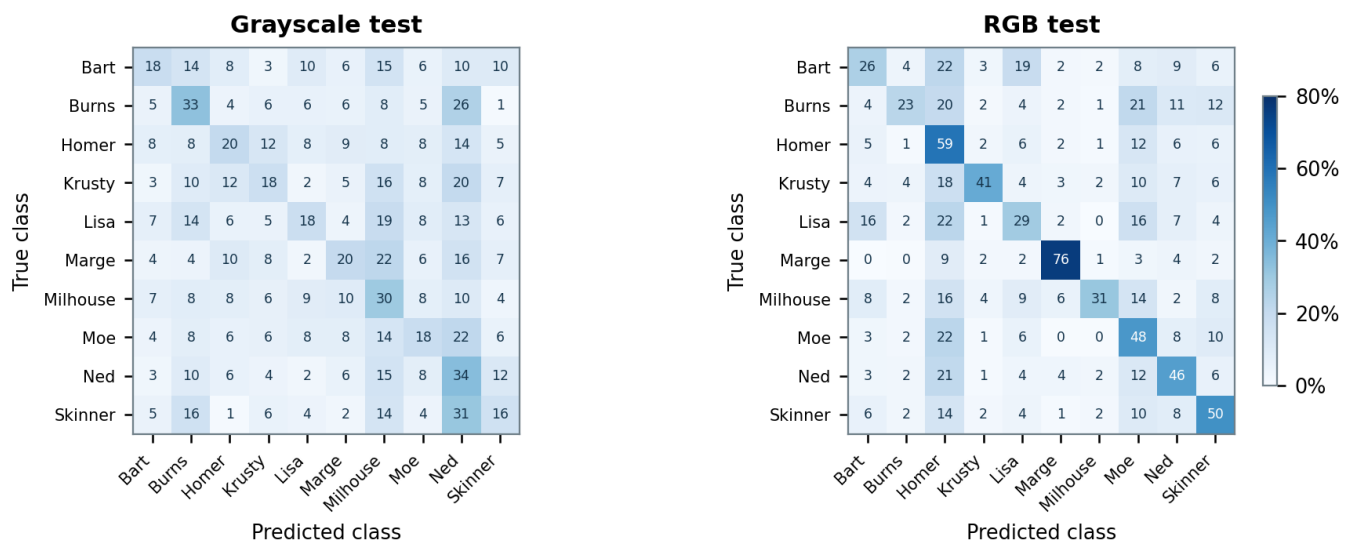
The lowest tested learning rate won for both modalities. RGB was consistently stronger, but normalization interacted with modality: z-score was best for grayscale, whereas simple 0-1 scaling was best for RGB.

5 Final validation and test performance

The selected models were evaluated with accuracy, macro-averaged precision, recall and F1. Because every class has equal support (200 test images), macro and weighted recall equal overall accuracy. Test metrics were not used to choose hyperparameters.

Model / split	Accuracy	Macro precision	Macro recall	Macro F1
Grayscale / validation	26.19%	27.76%	26.19%	25.94%
Grayscale / test	22.65%	23.75%	22.65%	22.39%
RGB / validation	42.75%	48.11%	42.75%	42.81%
RGB / test	42.85%	48.60%	42.85%	43.04%

Row-normalized test confusion matrices (% of each true class)



Grayscale confusion pattern

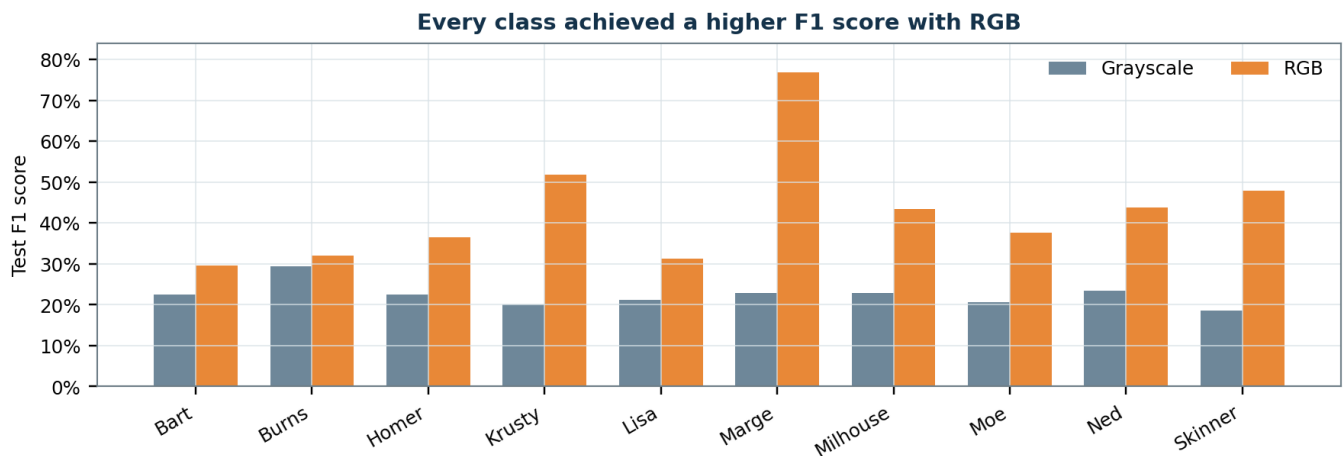
- Ned was over-predicted: Skinner -> Ned (31.0%), Burns -> Ned (25.5%), and Moe -> Ned (21.5%).
- Ned had the highest grayscale recall (34.5%) but low precision (17.65%).

RGB confusion pattern

- Homer attracted many errors: Moe -> Homer and Lisa -> Homer (22.5% each), Bart -> Homer (22.0%).
- Marge was the clearest RGB class, with 76.0% recall and 77.55% precision.

6 What colour information contributed

RGB improved test accuracy from 22.65% to 42.85%, a gain of 20.20 percentage points. Macro F1 increased by 20.65 points, and every character achieved a higher F1 score. Therefore, the improvement is not caused by only one unusually easy class.



Character	Gray F1	RGB F1	Gain
Marge	22.86%	76.77%	+53.91 pp
Krusty	20.00%	51.74%	+31.74 pp
Skinner	18.55%	47.85%	+29.30 pp
Milhouse	22.94%	43.36%	+20.42 pp
Ned	23.35%	43.75%	+20.40 pp

Interpretation

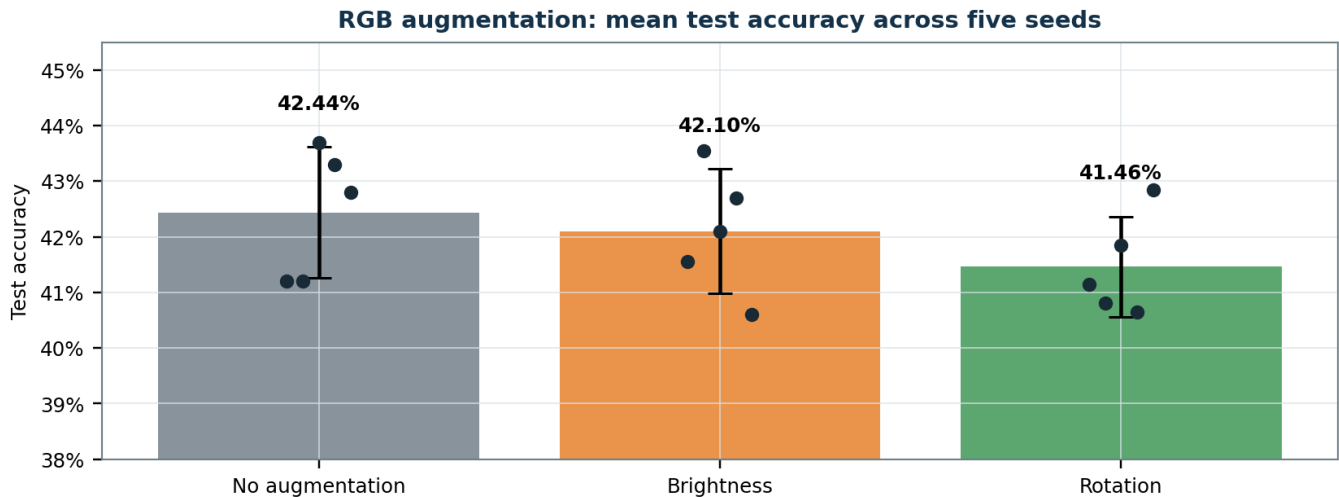
- Marge's distinctive blue hair provides a plausible colour cue, matching the largest gain.
- Krusty, Skinner, Milhouse and Ned also benefited strongly from RGB channel patterns.
- Burns gained the least in F1 (+2.73 pp): RGB precision rose, but recall fell from 33% to 23%.

Does grayscale contain enough information?

Grayscale retains edges, silhouettes and intensity, so performance remained above the 10% random baseline. However, 22.65% test accuracy and uniformly low class F1 scores show that shape and brightness alone were insufficient for reliable separation by a linear classifier. RGB adds informative channel-specific patterns, but its 42.85% accuracy also shows that colour alone does not solve the problem: pose, background and visual similarity still create substantial overlap.

7 RGB data augmentation

The selected RGB configuration was retrained under three conditions: no augmentation, one random brightness copy per image, and one random rotation copy per image. Augmentation occurred on raw images before flattening and normalization, doubling the training set to 12,800 examples. Each condition used five training-order seeds (11, 22, 33, 44, 55).



Condition	Training examples	Mean test accuracy	Sample SD	vs. baseline
No augmentation	6,400	42.44%	1.18%	-
Brightness	12,800	42.10%	1.12%	-0.34 pp
Rotation	12,800	41.46%	0.90%	-0.98 pp

Interpretation. Neither augmentation improved the mean test result, and the error bars overlap. Brightness changes may be largely redundant after 0-1 scaling. Rotations move important pixels to different flattened positions and introduce interpolation or border artefacts; a linear perceptron has no built-in translation or rotation invariance. These results do not show that augmentation is generally harmful - only that these transformations did not help this representation and model under the tested ranges.

8 Conclusions and reproducibility

Main conclusions

- RGB substantially outperformed grayscale.
- Low learning rates were most reliable for both modalities.
- Colour helped all classes, especially Marge and Krusty.
- The unaugmented RGB baseline had the highest five-seed mean.

Reproducibility and limits

- Fixed split seed 42; augmentation seed 2026.
- All preprocessing statistics came from training data only.
- The final model is linear; nonlinear or spatial models may separate the characters better.
- Five runs quantify seed variability, but no significance test was performed.